COMSATS University Islamabad, Lahore Campus

Midterm Examination – SPRING 2025

Course Title:	Design and Analysis of Algorithms	Course Code:	CSC301	Credit Hours:	3(3,0)
Course Instructor:	Dr Atif Saeed	Programme Name:	BSCS-BSSE		
Semester:	5th	Batch:	FA23-BCS-BSE	Section:	A, B, C, D
Date:	16-04-2025				
Time Allowed:	90 Minutes	Maximum Marks:	50		
Student's Name:	[redacted]	Reg. No.	[redacted]		

Important Instruction Guidelines:

- Be precise and to the point while answering any question.
- Show all your work, as partial credits will be given. You will be graded not only on the correctness of your answer, but also on the clarity with which you express it. Please be neat.
- Good luck!

Question 1: CLO: <1>; Bloom Taxonomy Level: <Applying> [Marks: 02 + 03 = 05]

(a) Why is asymptotic notation crucial in the analysis of algorithms? Provide two reasons.
(b) List two drawbacks of benchmarking as an empirical method.

Question 2: CLO: <2>; Bloom Taxonomy Level: <Analyzing> [Marks: 10]

Prove or disprove: $\frac{n^2+10\ n\log n}{n^2+n} \in \Theta(1)$

Question 3: CLO: <2>; Bloom Taxonomy Level: <Analyzing> [Marks: 10]

For the following code snippet, provide a line-by-line analysis and construct function T(*n*) that give the runtime of this code snippet as a function of "*n*". Also determine the Big-Oh of this code snippet.

```
a. for (int i = 1; i <= n; i *= 2)
b.     for (int j = 0; j < i; ++j)
           result += compute(i, j);

c. int compute(int a, int b) {
d. if (b == 0) return 1;
e. return compute(a, b - 1) + a;
f. }
```

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Owner of SuperCUI

Question 4: **CLO: <2>; Bloom Taxonomy Level: <Analyzing>** **[Marks: 3.5 + 3.5 + 3 = 10]**

In not more than five lines each, provide the answers to the following questions. Be reasonable in your assumptions. Your answer must appear in the first few words, followed by strong justification to support your answer.

a. You are implementing a high-frequency trading leaderboard where updates happen in milliseconds. Would **Quick Sort** or **Merge Sort** be more suitable for maintaining rankings? Justify your answer.

b. A database frequently retrieves records falling within specific income brackets. Would you recommend **Merge Sort** or **Quick Sort** for initial data organization before enabling range queries? Explain.

c. List two practical limitations of Merge Sort?

Question 5: **CLO: <2>; Bloom Taxonomy Level: <Analyzing>** **[Marks: 10]**

Solve the following recurrence using the "Recursion Tree Method".

$$T(n) = 2T\left(\frac{n}{3}\right) + T\left(\frac{n}{4}\right) + n$$

Question 6: **CLO: <2>; Bloom Taxonomy Level: <Analyzing>** **[Marks: 5]**

Solve the following recurrence using the "Master Theorem Method".

$$T(n) = 8T\left(\frac{n}{2}\right) + n^2$$

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